

Fig. 3: Frequency response of low-pass filter with $N=49$, $f_c=1.2$

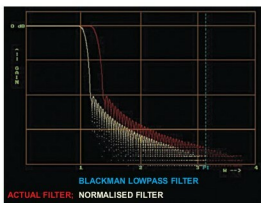


Fig. 5: Frequency response of low-pass filter with $N=149$, $f_c=1.2$

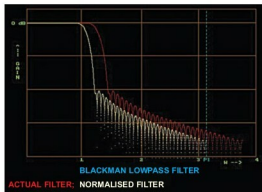


Fig. 4: Frequency response of low-pass filter with $N=99$, $f_c=1.2$

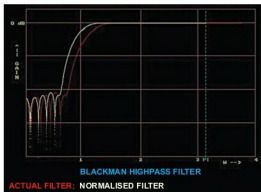


Fig. 6: Frequency response of high-pass filter with $N=45$, $f_c=1.2$

observe the effect of N on frequency response. All values of $|H(j\omega)|$ are in decibels. A normalised frequency scale is used.

Low-pass filter. Simulation of low-pass filter using Blackman window is done through the software. Figs 3 to 5 are the outputs of the low-pass filter for different window length of $N=49$, 99 and 149, respectively.

From Figs 3 to 5, we have following observations when N increases from 49 to 149 at cutoff frequency (f_c)=1.2:

1. The widths of the pass band and side lobes decrease
2. The slope of the transition band increases, that is, the band develops a sharper cutoff
3. As N increases, the number of ripples increase in the pass band

4. The amplitude and shape of the end-ripple remains the same but it gets shifted towards the edge and finally out of the pass band for larger values of N

5. As N increases, the ripples in the pass band get smoothed out. A key difference in the observed responses of the rectangular is the absence of the Gibbs phenomenon. This owes itself to the smoother shape of the Blackman window in the time-domain, with the notable absence of a discontinuity that would lead to the Gibbs phenomenon.

6. As N increases the widths of the lobes decrease in the stop band

Other frequencies

In the next section we present the outputs of high-pass, band-reject and band-pass filters.

Frequency response of high-pass filter with $N=45$, $f_c=1.2$ is shown in Fig. 6. Frequency response of band-reject filter with $N=67$, with lower cutoff frequency (f_l)=1.2 and upper cutoff

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